

EC 504
Fall, 2025
Homework 4

Due Tuesday, November 1, 11:59 pm on your SCC account

Problem 4.1 (20 pts)

Below are a set of quick true false type questions on graphs. Please provide brief answers.

- (a) An undirected graph with no cycles is a tree.
- (b) Consider an undirected graph where, for every pair of nodes i, j , there exists a unique simple path between them. This graph must be a tree.
- (c) Kruskal's algorithm for finding a minimum spanning tree grows a single tree into a minimum spanning tree.
- (d) The worst case complexity of the fastest minimum spanning tree algorithm is $O(n \log n)$, where n is the number of vertices in the graph.
- (e) Consider a minimum spanning tree T in a connected undirected graph (V, E) . Assume that the edge weights in the graph are all distinct (e.g. no two edges have the same weight). Then, an edge i, j in T must have the property that it is either the minimum weight edge connected to vertex i or the minimum weight edge connected to vertex j .
- (f) Vertices v_1 and v_2 are neighbors in an undirected graph if and only if there is a path between them.
- (g) Prim's algorithm and Kruskal's algorithm can also be used to find the maximum weight spanning tree in a graph.
- (h) Consider a minimum spanning tree in an undirected graph $G = (V, E)$ where all the edge weights are distinct. Then, if an edge e has the largest weight of all the edges in a cycle, that edge e cannot be part of a minimum spanning tree.
- (i) If a weighted, undirected graph has unique weights on its edges, the smallest weight edge on every cycle must be part of the minimum spanning tree.
- (j) If you add an extra edge to a minimum spanning tree, you form a graph with exactly one cycle.

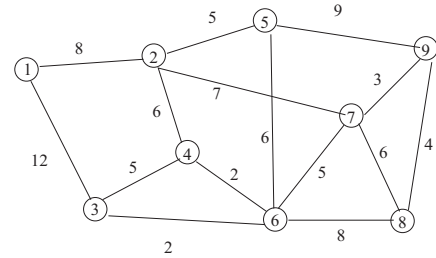
Problem 4.2 (20 pts)

Consider an undirected graph $G = (V, E)$.

- (a) Describe an $O(|V|)$ -time algorithm for determining whether the graph contains a cycle. .
- (b) Describe an algorithm to find the minimum number of edges that must be removed from a given undirected graph in order to make it acyclic. Note the graph may not be connected.

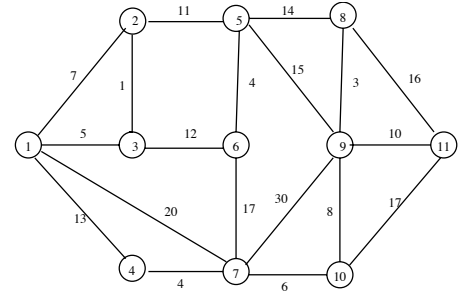
Problem 4.3 (10 pts)

Consider the undirected, weighted graph on the right. List the order in which the nodes will be visited in a Breadth-First Search, starting from node 5. Assume that the arcs out of a node will be visited in clockwise order, starting from straight up direction. Show the tree which is generated by the relation that node i 's parent is the node which puts node i into the queue.



Problem 4.4 (10 pts)

For the graph on the right, compute the minimum spanning tree using Kruskal's algorithm. Sort the edge weights, and show the partial forest of trees that occur in Kruskal's algorithm after **inserting** every 4 edges. Compute the total weight of the minimum spanning tree.



Problem 4.5 (20 pts)

Consider a path between two vertices s and t in an undirected, *connected* weighted graph G where no two edges have the same weight. The bottleneck length of this path is the maximum weight of any edge in the path. The bottleneck distance between s and t is the minimum bottleneck length of any path from s to t . Describe an algorithm to compute the bottleneck distance between every pair of vertices in an arbitrary undirected weighted graph. (Hint: Minimum spanning trees will help...)

Problem 4.6 (10 pts)

Consider the undirected weighted graph in the right. Show each step of Prim's algorithm for finding a minimum spanning tree in the graph.

